
Alternativa

a) (1 punt) Calcula $\int \left(2\sqrt{x} - \frac{1}{\sqrt{x}}\right) dx$ i $\int \frac{1}{\sqrt[3]{x}} dx$.

Solució. Escrivint les arrels com a potències, $2x^{1/2} - x^{-1/2}$ i $x^{-1/3}$:

$$\int \left(2x^{1/2} - x^{-1/2}\right) dx = 2 \frac{x^{3/2}}{3/2} - \frac{x^{1/2}}{1/2} + k = \frac{4}{3}x\sqrt{x} - 2\sqrt{x} + k, \quad \int x^{-1/3} dx = \frac{x^{2/3}}{2/3} + k = \frac{3}{2}\sqrt[3]{x^2} + k.$$